

MATH3026/4022 Lecture 10

Non-stationary and Seasonal Time Series

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- ▶ Recap on stationarity and removal of stationarity through differencing
- ▶ The autoregressive integrated moving average (ARIMA) model
- ▶ Non-constant variance and transformations
- ▶ Identifying seasonality
- ▶ The seasonal ARIMA model

Non-stationary and Seasonal Time Series

During previous lectures, we have focused mainly on stationary time series and how we can fit stationary models to time series data.

In reality, many time series are not stationary.

In this lecture, we'll begin to explore how we might model time series data that are non-stationary or that contain seasonal trends.

Assessing Stationarity - Recap

From Lecture 2, we recall that a time series is **stationary** if

- ▶ It contains no trend (i.e. no systematic change in mean over time).
- ▶ It contains no systematic change in variance over time.
- ▶ Strictly periodic variations (i.e. seasonality) have been removed.

Assessing Stationarity - Recap

In practice, we consider **weak stationarity** (and if the term 'stationary' is used with respect to a time series, we assume this implies that the time series is weakly stationary).

A time series $\{X_t : t \in \mathbb{Z}\}$ is said to be **weakly stationary** if

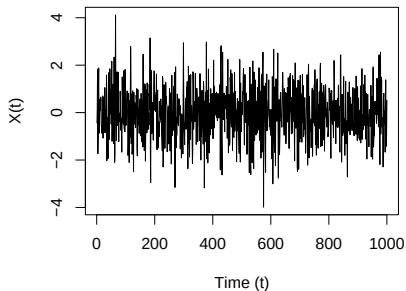
- (i) $\mu(t)$ is independent of t .
- (ii) $\text{Var}(X_t) = \sigma_X^2(t) = \sigma^2 < \infty$ for all t .
- (iii) $\gamma(t, t+h)$ is independent of t for each $h \in \mathbb{Z}$.

Assessing Stationarity

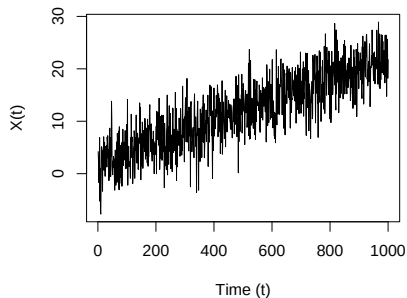
As such (and obviously) we would want to check whether or not a time series appears to be stationary before considering non-stationary models.

A time plot would allow us to assess whether or not the mean appears to be constant.

Stationary series



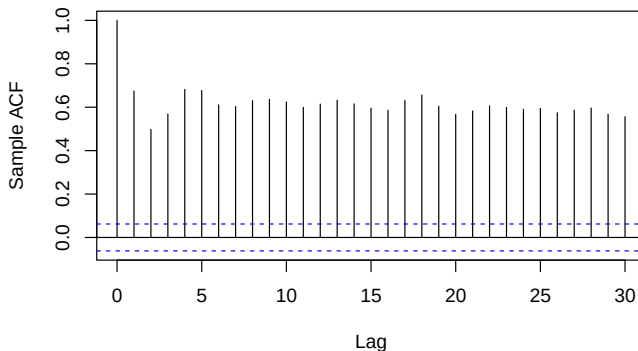
Non-stationary series



Assessing Stationarity

Usually, the sample ACF values for a non-stationary time series do not decay (or decay very slowly) as the lag increases.

An example plot of the sample ACF from a non-stationary time series is shown below.



Removing non-stationarity: Differencing

A standard approach to take when a time series is non-stationary is to take differences until the (differenced) process appears to be stationary.

Assuming that $\mathbf{x} = (x_1, \dots, x_n)^\top$ is an observed set of time series data, we might consider the first or second differences

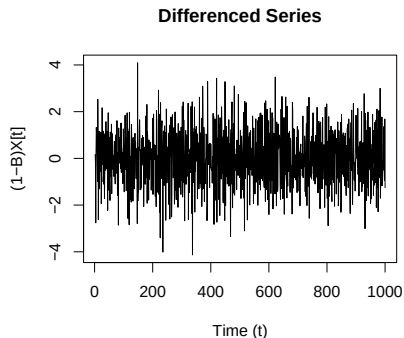
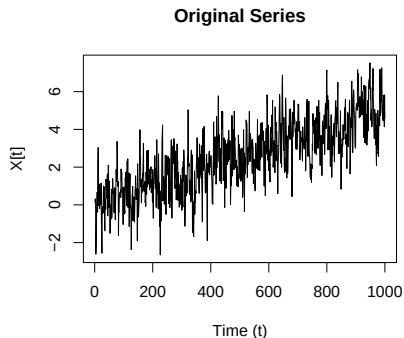
$$\begin{aligned}\nabla x_t &= (1 - B)x_t = x_t - x_{t-1} \\ \nabla^2 x_t &= \nabla (\nabla x_t) = (1 - B)^2 x_t \\ &= x_t - 2x_{t-1} + x_{t-2}.\end{aligned}$$

We note that we should stop differencing as soon as the process appears to be stationary (from the time plot) and we should not over-difference.

Typically, we consider $\nabla^d x_t$ only for $d = 0, 1$ and 2 .

Removing non-stationarity: Differencing

The time plots below show observations from a time series with a linear trend (original series) and a similar plot for the first difference of the series.

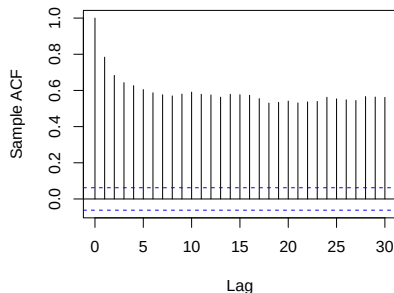


We see that the differenced series appears to be stationary.

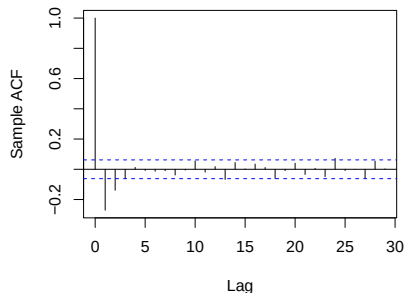
Removing non-stationarity: Differencing

To double-check, we also produce plots of the sample ACFs for the original series and the differenced series.

Original Series



Differenced Series



The ARIMA Model

We can fit the types of models seen in earlier lectures to time series where differencing has been used to make the series stationary.

The **autoregressive integrated moving average** (ARIMA) model is defined as follows:

Suppose X_t is a time series and W_t is defined

$$W_t = \nabla^d X_t = (1 - B)^d X_t.$$

The autoregressive integrated moving average with order (p, d, q) is written

$$W_t = \phi_1 W_{t-1} + \dots + \phi_p W_{t-p} + Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_q.$$

We refer to this as an *ARIMA*(p, d, q) model.

The ARIMA Model

We see that the $ARIMA(p, d, q)$ model is simply an $ARMA(p, q)$ model fitted to a series where differencing has taken place d times.

If W_t is $(1 - B)^d X_t$ then an $ARIMA(p, d, q)$ model fitted to the series $\{X_t\}$ implies that an $ARMA(p, q)$ model is fitted to the differenced series $\{W_t\}$.

Using notation from earlier lectures, we may write

$$\begin{aligned} W_t - \phi_1 W_{t-1} - \dots - \phi_p W_{t-p} &= Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q} \\ \implies (1 - \phi_1 B - \dots - \phi_p B^p) W_t &= (1 + \theta_1 B + \dots + \theta_q B^q) Z_t \\ &\implies \phi(B) W_t = \theta(B) Z_t \\ &\implies \phi(B) [(1 - B)^d X_t] = \theta(B) Z_t. \end{aligned}$$

Non-constant Variance

Whilst differencing can be helpful where a time series has a non-constant mean, we should consider other approaches if a time series has a non-constant variance.

To identify whether or not a series has a non-constant variance, we might partition the observed series data into K subsets and calculate the sample mean ($\hat{\mu}_k$) and sample variance (s_k^2) for each subset $k \in \{1, \dots, K\}$.

Then, we produce two plots

- (i) A scatter plot of s_k^2 against $\hat{\mu}_k^2$.

If this plot indicates that s_k^2 is proportional to $\hat{\mu}_k^2$ then we might use a log transform on $\mathbf{x}$ and fit a model to the transformed series $\mathbf{y} = \log(\mathbf{x})$.

Non-constant Variance

(ii) A scatter plot of s_k^2 against $\hat{\mu}_k$.

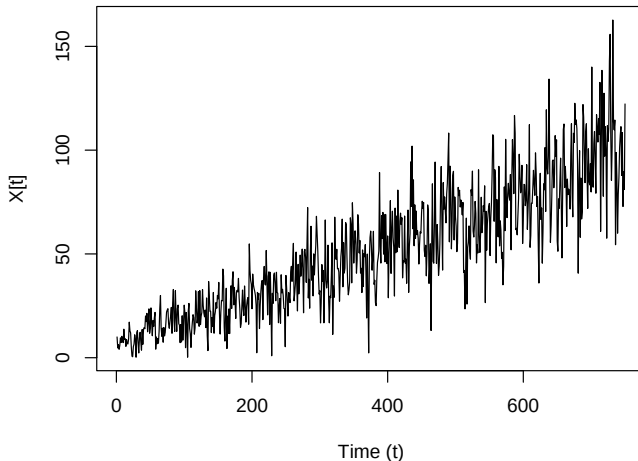
If this plot indicates that s_k^2 is proportional to $\hat{\mu}_k$ then we might use a square root transform on $\mathbf{x}$ and fit a model to the transformed series $\mathbf{y} = \sqrt{\mathbf{x}}$.

We note that each of these transformations is only appropriate where each value of $\mathbf{x}$ is greater than 0 (or ≥ 0 for the square root transformation).

We can also produce similar plots using running sample means and variances, where sample quantities are calculated for overlapping intervals of length n along the time series.

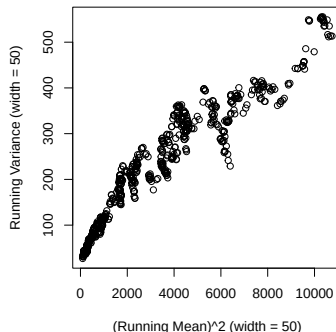
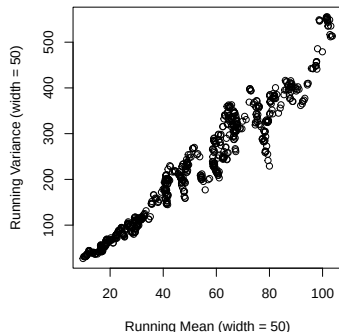
Non-constant Variance

As an example, consider the following time plot of time series data with a non-constant variance.



Non-constant Variance

Below are plots of the running mean and square of the running mean against the sample variance.



Non-constant Variance

We used a window width of $n = 50$ for calculations here.

We see that the left hand plot of s_k^2 against $\hat{\mu}_k$ appears to be more linear than that of s_k^2 against $\hat{\mu}_k^2$ and so we might consider a square root transformation for this set of data to account for non-constant variance.

Typically, we consider transformations only for **major** departures from a constant variance and often after differencing has been performed.

Seasonal Time Series

Next, we turn our attention to time series that exhibit a seasonal pattern.

We can think of many examples where observations may differ amongst time points in a periodic manner. For example

- ▶ Airline passenger numbers – typically these may be high during holiday periods (e.g. summer, Christmas etc.) but lower in winter months.
- ▶ Monthly temperature data for Nottingham – temperatures will vary from cold (winter months) to warm (summer months). It is likely that temperatures in January, February etc. for different years will be correlated.

Seasonal Time Series

If there is seasonality in an observed time series $\mathbf{x} = (x_1, \dots, x_n)^\top$ then the process may not be stationary.

If the seasonality appears to occur with a period h then we might difference the data at the seasonal lag h and remove the seasonal trend prior to modelling.

Whilst this approach might be satisfactory for removing seasonality in the mean, a series may also exhibit seasonality in the autocorrelation structure.

To account for seasonality in the autocorrelation structure, a model known as the **seasonal** ARIMA model (or *SARIMA* model) has been developed, which we'll outline later during this lecture.

Identifying Non-stationarity/Seasonality

To identify possible non-stationarity/seasonality we first produce a time plot of the series and look for trends (e.g. linear, quadratic), non-constant variance or regular cycles (possible non-stationarity).

We then look at a plot of the sample ACF for the data

1. If the sample ACF decays to zero fairly quickly then there is probably no need for differencing.

If not, we might consider differencing the time series.

2. If the sample ACF decays to zero quickly *but* there is cyclic structure (e.g. peaks at 12, 24, 36 and troughs at 6, 18, 30) then:
 - ▶ If the spikes die down quickly then we might not difference even though there may be seasonality in the data.
 - ▶ If the spikes don't die down quickly the series may not be stationary and we might like to difference the series by one season.

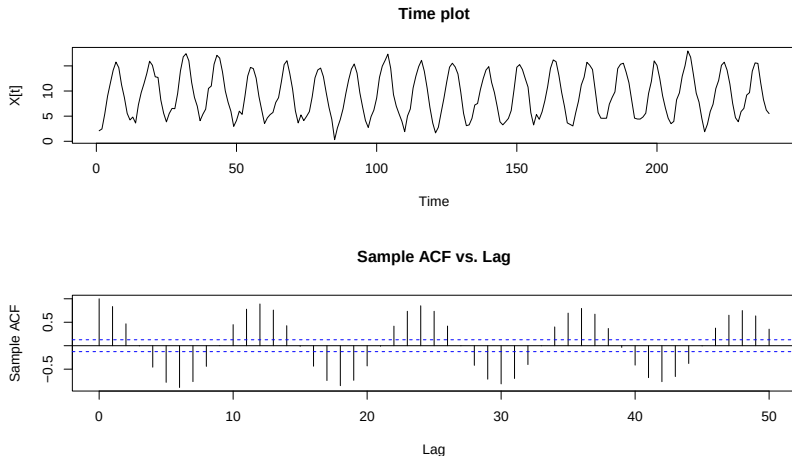
Identifying Non-stationarity/Seasonality

For example, if there is seasonality and the data are collected monthly, then we would difference by 12, defining

$$\begin{aligned}w_t &= x_t - x_{t-12} \\&= (1 - B^{12})x_t \\&= \nabla_{12}x_t.\end{aligned}$$

Then we should look at the sample ACF and time plot of w_t and decide whether or w_t is a stationary process.

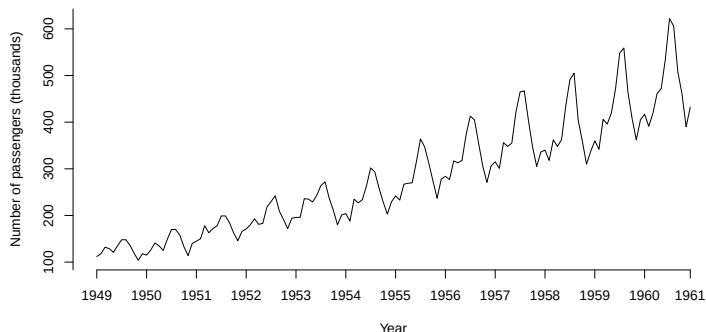
Example 10.1



We see that the ACF does not decay quickly and, instead, clearly shows peaks at 12, 24, 36, ..., with troughs at 6, 18, 30,

Example 10.2 - The US Air Passenger Data

From Lecture 2, recall the US Monthly Air Passenger Data from 1949–1960 (time plot shown below).



Clearly, this time series is non-stationary and exhibits seasonality.

Example 10.2 - The US Air Passenger Data

As the data have been collected monthly – and using our intuition with regard to seasonal travel trends over a year – we'd first difference the data with lag 12.

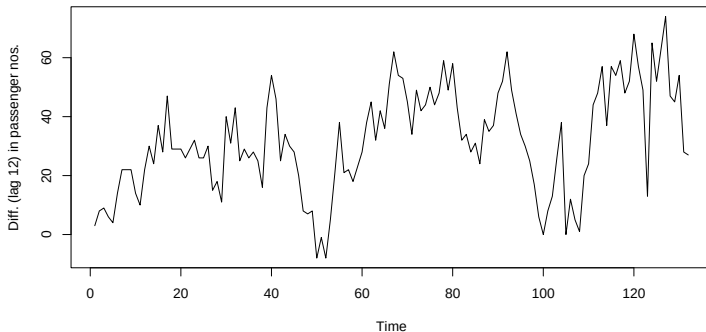
That is, if $\{X_t\}$ denotes our original time series, we define the lag 12 difference time series, $\{Y_t\}$ as

$$\begin{aligned} Y_t &= X_t - X_{t-12} \\ &= \nabla_{12} X_t \end{aligned}$$

Now we'll look at a time plot of the seasonally differenced time series.

Example 10.2 - The US Air Passenger Data

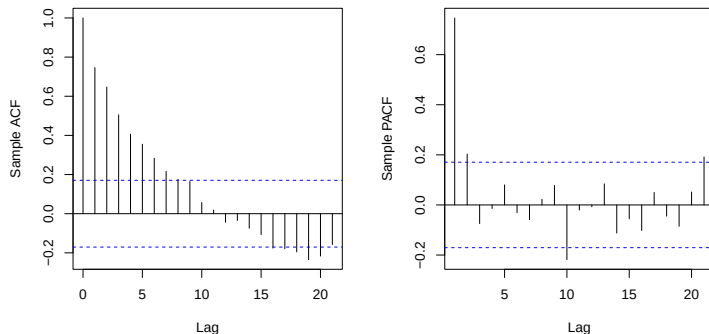
Below is a time plot of the seasonally differenced (at lag 12) US air passenger data



Although the seasonality appears to have been removed, this time series is not necessarily stationary.

Example 10.2 - The US Air Passenger Data

Below are plots of the sample ACF and sample PACF against the lag for the seasonally differenced data.



These plots suggest that the seasonally differenced data are not stationary.

Example 10.2 - The US Air Passenger Data

Finally, we'll take the **first difference** of the seasonally differenced US Air Passenger data.

So where $\{X_t\}$ is the original series and $\{Y_t\}$ is the seasonally differenced series (at lag 12), such that

$$Y_t = X_t - X_{t-12}.$$

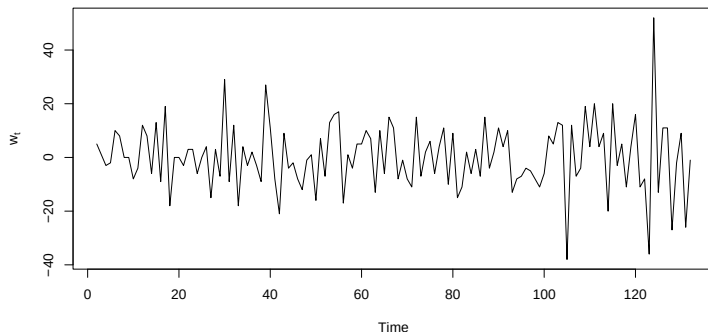
We take the first difference of the series $\{Y_t\}$, written

$$W_t = Y_t - Y_{t-1}.$$

Now, we examine a time plot for the series $\{W_t\}$.

Example 10.2 - The US Air Passenger Data

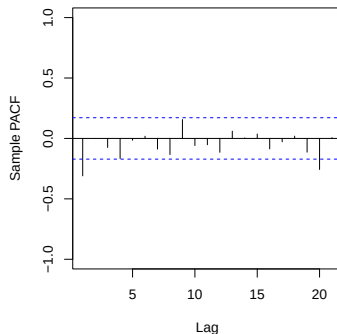
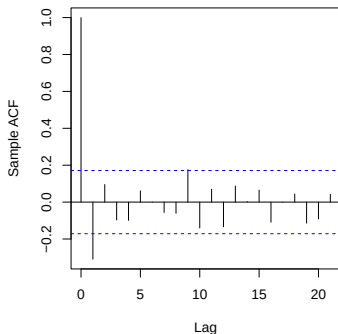
A time plot for the series $\{W_t\}$ is shown below.



These data appear to be fairly stationary and certainly the type of observed data to which we would consider fitting an *ARMA* model.

Example 10.2 - The US Air Passenger Data

Below are plots of the sample ACF and sample PACF against the lag for the series $\{W_t\}$



These plots suggest that the series $\{W_t\}$ is stationary and, hence, we would consider fitting an ARMA model using the observed values of this series.

Seasonal ARIMA (SARIMA) Model

The seasonal ARIMA (or SARIMA) model has been developed as an approach to modelling time series that contain seasonality and/or non-stationarity.

Suppose that $\{X_t : t \in \mathbb{Z}\}$ is a time series that contains seasonality.

Box and Jenkins (1970) proposed the following seasonal ARIMA filter:

$$\begin{aligned}\Phi_P(B^h)X_t &= \Theta_Q(B^h)W_t \\ \implies W_t &= \left[\Theta_Q(B^h)\right]^{-1} \Phi_P(B^h)X_t\end{aligned}\quad (10.1)$$

where $\Phi_P(\cdot)$ and $\Theta_Q(\cdot)$ are polynomials of orders P and Q , respectively.

Here, the filter is applied to $\{X_t\}$ to remove seasonality.

Seasonal ARIMA (SARIMA) Model

Since the filter is applied to $\{X_t\}$ to remove seasonality it follows that $\{W_t\}$ is a non-seasonal time series.

Suppose now that we fit a regular $ARMA(p, q)$ model to W_t , implying

$$\phi_p(B)W_t = \theta_q(B)Z_t \quad (10.2)$$

where $\phi_p(\cdot)$ and $\theta_q(\cdot)$ are polynomials of orders p and q , respectively.

Then, if we substitute W_t from (10.1) into (10.2), we obtain

$$\begin{aligned} \phi_p(B) \left[\Theta_Q(B^h) \right]^{-1} \Phi_P(B^h) X_t &= \theta_q(B) Z_t \\ \implies \phi_p(B) \Phi_P(B^h) X_t &= \theta_q(B) \Theta_Q(B^h) Z_t \end{aligned}$$

Seasonal ARIMA (SARIMA) Model

The model

$$\phi_p(B)\Phi_P(B^h)X_t = \theta_q(B)\Theta_Q(B^h)Z_t$$

is a type of **seasonal** ARIMA (or SARIMA) model of order

$$(p, 0, q) \times (P, 0, Q)_h.$$

The seasonal components are $\Phi_P(B^h)$ and $\Theta_Q(B^h)$.

The non-seasonal components are $\phi_p(B)$ and $\theta_q(B)$.

Seasonal ARIMA (SARIMA) Model

More generally, if $\{X_t\}$ is also differenced at lag h , D times, to remove non-stationarity because of the seasonal component and then differenced d times to remove non-stationarity because of the non-seasonal component, then, more generally, we refer to the SARIMA model as an

$$ARIMA(p, d, q) \times (P, D, Q)_h$$

model, where

(p, d, q) refers to the **non-seasonal** part of the model;
 $(P, D, Q)_h$ refers to the **seasonal** part of the model.

Seasonal ARIMA (SARIMA) Model

The general seasonal ARIMA model of order $(p, d, q) \times (P, D, Q)_h$ is written

$$\phi_p(B)\Phi_P(B^h)\nabla^d\nabla_h^D X_t = \theta_q(B)\Theta_Q(B^h)Z_t$$

or

$$\phi_p(B)\Phi_P(B^h)(1-B)^d(1-B^h)^D X_t = \theta_q(B)\Theta_Q(B^h)Z_t.$$

Here

$$\begin{aligned}\phi_p(B) &= 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p; \\ \Phi_P(B^h) &= 1 - \Phi_1 B^h - \Phi_2 B^{2h} - \dots - \Phi_P B^{Ph}; \\ \theta_q(B) &= 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q; \\ \Theta_Q(B) &= 1 + \Theta_1 B^h + \Theta_2 B^{2h} + \dots + \Theta_Q B^{Qh}.\end{aligned}$$

Example 10.3

Write out the form of the $ARIMA(0, 0, 0) \times (0, 1, 1)_{12}$ model for the time series $\{X_t\}$.

Example 10.4

Write out the form of the $ARIMA(0, 0, 1) \times (1, 0, 1)_{12}$ model for the time series $\{X_t\}$.

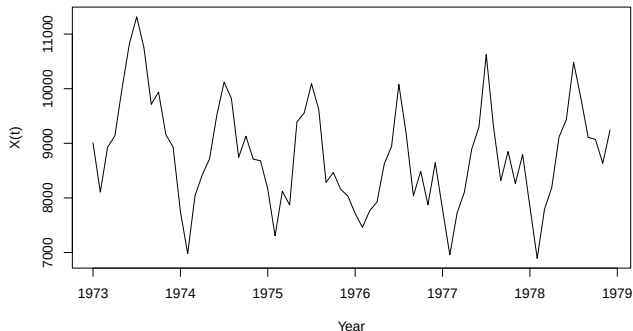
Example 10.5

Write out the form of the $ARIMA(0, 1, 1) \times (0, 1, 1)_4$ model for the time series $\{X_t\}$.

Inference for SARIMA models

SARIMA models can be fitted in R using the `arima` command, with the seasonal part of the model specified using the option `seasonal`.

Below is a time plot of the monthly numbers of accidental deaths in the USA for the years 1973 to 1978.



Inference for SARIMA models

This series shows clear seasonality and, as such, we'll fit a SARIMA model of the form

$$ARIMA(0, 1, 1) \times (0, 1, 1)_{12}$$

to these data.

If $\{X_t\}$ is the time series of the monthly number of accidental deaths, then this model is written

$$\begin{aligned}\phi_0(B)\Phi_0(B^{12})(1-B)(1-B^{12})X_t &= \theta_1(B)\Theta_1(B^{12})Z_t \\ \implies (1-B)(1-B^{12})X_t &= (1+\theta_1B)(1+\Theta_1B^{12})Z_t \\ \implies (1-B-B^{12}+B^{13})X_t &= (1+\theta_1B+\Theta_1B^{12}+\theta_1\Theta_1B^{13})Z_t \\ \implies (1-B-B^{12}+B^{13})X_t &= Z_t + \theta_1Z_{t-1} + \Theta_1Z_{t-12} + \theta_1\Theta_1Z_{t-13}\end{aligned}$$

We have three parameters to estimate: θ_1 , Θ_1 and σ_z^2 .

Inference for SARIMA models

The R command to fit the model is

```
model<-arima(USAccDeaths,order=c(0,1,1),  
             seasonal=list(order=c(0,1,1),period=12),  
             method="ML")
```

The fitted model output is

Coefficients:

	ma1	sma1
	-0.4303	-0.5528
s.e.	0.1228	0.1784

σ^2 estimated as 99347: log likelihood = -425.44,
aic = 856.88

Here

- ▶ `ma1`: denotes the estimate of θ_1 and its associated standard error.
- ▶ `sma1`: denotes the estimate of Θ_1 and its associated standard error.
- ▶ `sigma^2`: denotes the estimate of σ_z^2 .

Because the fit is likelihood-based (we assume that the white noise terms are normally distributed), it is worth noting that our asymptotic arguments and results with regard to maximum likelihood estimators will hold for SARIMA models.

Lecture 10: Learning Outcomes

- ▶ Know what is meant by a stationary time series and understand how to assess whether or not a time series is stationary.
- ▶ Understand how to remove non-stationarity using differencing.
- ▶ Understand how to define and interpret the $ARIMA(p, d, q)$ model.
- ▶ Know how to use plots of sample mean and sample variance to identify non-constant variance and understand that transformations can be used to remove non-constant variability.
- ▶ Understand what is meant by seasonality and be able to define and interpret seasonal ARIMA (SARIMA) models.